

# Force & Motion

## Newton's 3rd Law

Every Action Has an Equal and Opposite Reaction

### What is Newton's 3rd Law?

*"For every action force, there is an equal and opposite reaction force."*

This means whenever one object pushes or pulls on another, the second object pushes or pulls back with the **same strength** but in the **opposite direction**.

### Part 1 — Key Ideas

| Concept            | Explanation                                         | Key Point                |
|--------------------|-----------------------------------------------------|--------------------------|
| Action Force       | The first push or pull exerted on an object         | Always causes a reaction |
| Reaction Force     | The equal and opposite push/pull that comes back    | Always equal in size     |
| Equal Magnitude    | Both forces are exactly the same size/strength      | Same number of Newtons   |
| Opposite Direction | Forces act in exactly opposite directions           | 180° apart               |
| Different Objects  | Action and reaction forces act on DIFFERENT objects | Never cancel each other! |

### Part 2 — Real-World Examples



#### Rocket Launch

Rocket engines push hot gas BACKWARD (action). The gas pushes the rocket FORWARD (reaction). This is why rockets can travel in space — no ground needed!

|                        |                                                                                                                                                       |
|------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|
| ■ <b>Swimming</b>      | A swimmer pushes water BACKWARD with their hands (action). The water pushes the swimmer FORWARD (reaction). The harder they push, the faster they go! |
| ■ <b>Bird Flight</b>   | A bird's wing pushes air DOWNWARD (action). The air pushes the bird UPWARD (reaction). This lift force keeps birds in the sky!                        |
| ■ <b>Fish Swimming</b> | A fish pushes water BACKWARD with its tail (action). Water pushes the fish FORWARD (reaction). Same principle as swimming!                            |
| ■ <b>Walking</b>       | Your foot pushes BACKWARD on the ground (action). The ground pushes your foot FORWARD (reaction). Without friction, you couldn't walk!                |
| ■ <b>Archery</b>       | The bowstring pushes the arrow FORWARD (action). The arrow pushes the bowstring BACKWARD (reaction). Equal force, opposite directions!                |

### Part 3 — Important Notes & Common Mistakes

#### ■ Remember These!

- Action-reaction pairs ALWAYS act on DIFFERENT objects
- They CANNOT cancel each other out (different objects!)
- Both forces are ALWAYS equal in magnitude
- Both forces are ALWAYS opposite in direction
- Newton's 3rd Law applies to ALL forces — gravity, friction, normal force

#### ■ Common Mistakes

- Thinking action force is bigger than reaction force
- Thinking the forces cancel each other out
- Confusing 3rd law pairs with balanced forces
- Forgetting the forces act on DIFFERENT objects
- Thinking only moving objects experience 3rd law forces

### Part 4 — Newton's Three Laws at a Glance

| Law                      | Statement                                                                                                     | Example                                                            |
|--------------------------|---------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>1st Law (Inertia)</b> | An object at rest stays at rest. An object in motion stays in motion UNLESS acted on by a net external force. | A ball rolling keeps rolling until friction stops it.              |
| <b>2nd Law (F = ma)</b>  | Force equals mass × acceleration. Bigger force = bigger acceleration. More mass = less acceleration.          | Kicking a heavy ball vs. a light ball — the light one goes faster! |

|                                   |                                                                                              |                                                     |
|-----------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------|
| <b>3rd Law (Action-Re action)</b> | For every action, there is an equal and opposite reaction force acting on different objects. | Rocket pushes gas back → gas pushes rocket forward! |
|-----------------------------------|----------------------------------------------------------------------------------------------|-----------------------------------------------------|

## Part 5 — Practice Questions

|    | Question                                                                  | Answer                                                                                                    |
|----|---------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------|
| Q1 | State Newton's 3rd Law in your own words.                                 | <b>For every action force there is an equal and opposite reaction force acting on a different object.</b> |
| Q2 | When a rocket pushes gas backward, what happens to the rocket?            | <b>The gas pushes the rocket forward with an equal force (reaction force).</b>                            |
| Q3 | Why do action and reaction forces NOT cancel each other out?              | <b>They act on DIFFERENT objects — they cannot cancel as they are not on the same object.</b>             |
| Q4 | A swimmer pushes water backward. Describe the reaction force.             | <b>The water pushes the swimmer forward with equal force — this propels the swimmer.</b>                  |
| Q5 | Name 3 everyday examples of Newton's 3rd Law.                             | <b>Any 3 from: walking, swimming, rocket, bird flight, fish swimming, archery, rowing, jumping.</b>       |
| Q6 | If you push a wall with 50N force, how hard does the wall push back?      | <b>50N — equal and opposite!</b>                                                                          |
| Q7 | Identify the action and reaction in: "A bird pushes air downward to fly." | <b>Action: bird's wing pushes air downward.<br/>Reaction: air pushes bird upward.</b>                     |
| Q8 | How is Newton's 3rd Law different from balanced forces?                   | <b>3rd law pairs act on DIFFERENT objects; balanced forces act on the SAME object.</b>                    |

## Key Takeaways

- **3rd Law** Every action = equal & opposite reaction on a DIFFERENT object
- **Magnitude** Action and reaction forces are ALWAYS equal in size

**Direction**

They are ALWAYS opposite in direction

**Objects**

They act on DIFFERENT objects — never cancel out!

**Walking**

Foot pushes ground back → ground pushes you forward

**Swimming**

Push water back → water pushes you forward

**All Forces**

3rd law applies to ALL forces: gravity, friction, tension, etc.

**Remember: Physics is everywhere!**

Every step you take, every swim, every rocket launch — Newton's 3rd Law is working!